Debriefing – BestDeal Product Standardization

Project scope: Standardization of product quantities and units

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Version: 1.0

Status: Scope confirmed with client (see chosen focus below and Source section)

1. Introduction

BestDeal offers its members a procurement platform for purchasing non-product-related spend items from more than 50 major suppliers. Its assortment contains approximately 1.2 million products, ranging from office supplies and food products to technical products. These products are presented through the BestDeal webshop, which is powered by a Product Information Management (PIM) system. The PIM receives daily product feeds from individual suppliers; the dataset provided to us for this project is this live supplier data.

The challenge is that supplier product data is not always represented consistently. Even when different suppliers offer the same product, quantities, packaging units and units of measurement can differ. For example, one supplier may provide a value as 0.33 L while another provides 330 ml. Printing paper can also be represented and priced differently depending on the supplier. These differences can make product and price comparison in the webshop difficult or misleading.

2. Client

Client problem

BestDeal receives supplier-specific product feeds, meaning that the same product can contain different quantity and unit information depending on the supplier. Manually standardizing the entire assortment is economically unfeasible because of the size of the product range.

Client wants, needs and wishes

- A scalable method for standardizing product quantities and units across supplier feeds.
- More consistent information in the webshop so equivalent products can be compared more accurately.
- The original supplier-specific ordering requirements must remain available in the backend.
- The project should focus specifically on units and quantities rather than trying to standardize all product information.

Expected tangible outcome

We will investigate and develop an IT-based solution or prototype focused specifically on standardizing product units and quantities across supplier data. The outcome should demonstrate how standardized units and quantities can support accurate price comparison in the webshop while preserving the supplier-specific ordering requirements.

Project scope

In scope:

- Product quantity standardization
- Product unit standardization
- Analysis of relevant supplier data feeds

Out of scope:

- Product descriptions and images
- Cross-selling (not part of the selected assignment)
- Upselling (not part of the selected assignment)
- Seasonal product recommendations (not part of the selected assignment)

3. Context Analysis

The BestDeal PIM is fed daily by individual supplier product feeds. The EAN (European Article Number) acts as a common denominator that can help identify and compare the same product across suppliers. However, the quality and structure of the product information varies by supplier and product group.

The first stage of our project will therefore focus on understanding the supplied dataset before selecting a technical solution. We will investigate how quantities and units are represented, which inconsistencies occur most frequently, how EAN values can support matching, and whether data is missing, incorrect or unusually formatted.

During the client meeting it was also mentioned that one supplier folder may not be usable. We will identify this through dataset exploration rather than assuming in advance which supplier should be excluded.

Market and comparable solutions

The project will examine how e-commerce, procurement and Product Information Management environments approach product normalization, unit conversion, quantity normalization, supplier-feed integration and automated data-quality improvement. Comparable approaches can help determine which techniques are suitable for BestDeal's situation.

Potential users and target audience

The primary users are BestDeal employees responsible for product information, supplier feeds, the PIM and webshop data. BestDeal customers are indirect users because standardized product information can make supplier offers easier to understand and compare.

4. IT Analysis

Different technical approaches should be investigated for the agreed scope of unit and quantity standardization. Potential approaches include conventional data preprocessing, rule-based unit conversion, pattern recognition, machine learning or AI for inconsistent supplier fields, anomaly detection, and a hybrid combination of rules and AI.

Some standardization problems are deterministic. For example, converting 0.33 L to 330 ml can be handled using predefined conversion rules. More complex cases may involve inconsistent field structures, packaging information, missing values or supplier-specific formats. These cases should be analyzed to determine whether AI or machine learning adds value compared with conventional data-processing methods.

5. Problem Definition

BestDeal receives daily product data feeds from multiple suppliers. Although identical products can often be identified using their EAN, suppliers may represent quantities and units differently. These inconsistencies make accurate product and price comparison in the webshop difficult. Manual standardization is not economically feasible because BestDeal manages approximately 1.2 million products. BestDeal therefore needs a scalable method for identifying and standardizing supplier-specific quantity and unit information while preserving the original supplier ordering requirements.

6. Design Challenge / Main Research Question

How can BestDeal standardize units and quantities across different supplier data feeds so that prices for equivalent products can be accurately compared in the webshop?

7. Direction for the Project Plan

After the debriefing is validated with the client, it can be elaborated into the project plan. The project can then define sub-questions around data quality and inconsistencies, standardization requirements, existing approaches, appropriate IT/AI techniques, prototype development and evaluation. Goals, planning and project risks can subsequently be defined based on the agreed scope.

Source

C2Value - Project Case (2026 Spring), BestDeal case description and assignment; client meeting notes and our scope-confirmation email to the client.